

LLM-Augmented Decision Intelligence for Real-Time Supply Chain Risk Assessment: A Parameterized Simulation and FMEA-Based Framework

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Abstract

Modern supply chain systems operate in highly volatile environments where conventional risk management approaches are structurally insufficient to capture compound and multi-dimensional disruptions. This paper proposes a novel LLM-augmented decision intelligence framework that integrates a six-layer processing architecture with FMEA-based risk quantification and large language model contextual reasoning for real-time supply chain risk assessment. The proposed framework advances existing approaches by combining deterministic risk formalization—through the FMEA-based Risk Priority Number (RPN) scoring mechanism—with the contextual reasoning capabilities of large language models within a unified, modular enterprise pipeline. Empirical validation is conducted on two complementary datasets: a fully deterministic 120-scenario parameterized simulation enabling controlled stress-testing of compound risk interactions, and the publicly available DataCo Smart Supply Chain dataset comprising 180,519 transaction-level records, which constitutes the primary basis for performance claims and real-world generalizability assessment. On the DataCo dataset, the proposed framework achieves 96.8% recall—the operationally critical metric in high-stakes risk detection contexts where false negatives carry substantially higher cost than false alarms—with McNemar's test confirming statistically significant superiority over baseline systems ($\chi^2 = 156.4$, $p < 0.001$). Five-fold stratified cross-validation confirms highly stable performance (Recall: $96.9\% \pm 0.4\%$). A strong RPN-to-cost correlation ($r = 0.89$, $p < 0.001$) validates FMEA scoring as a financially grounded risk proxy. The framework is designed for enterprise integration via a REST API layer compatible with leading ERP and CRM platforms. These results collectively demonstrate that integrating deterministic FMEA quantification with LLM-based contextual reasoning yields a scalable, auditable, and operationally robust foundation for next-generation supply chain decision intelligence.

Keywords: *Supply chain risk management · Large language models · FMEA · Decision intelligence · Parameterized simulation · McNemar's test · Recall-optimized classification · Enterprise AI deployment · Ablation study · Non-linear risk modeling*

1. INTRODUCTION

The increasing complexity and volatility of global supply chain networks have exposed quantitatively significant limitations in conventional risk management paradigms. Static rule-based models and historical trend analysis are structurally incapable of capturing compound, multi-dimensional disruptions that propagate non-linearly across interconnected supply nodes [1], [2]. Disruption events such as the 2020–2021 global semiconductor shortage and post-pandemic logistics volatility underscored the operational cost of inadequate real-time risk intelligence, with industry estimates placing aggregate disruption costs at trillions of dollars annually [4], [12]. This operational imperative motivates the design of adaptive frameworks capable of real-time risk assessment, contextual anomaly detection, and auditable decision support.

Recent advances in large language models (LLMs) have demonstrated measurably strong capability for reasoning over unstructured and contextual enterprise data [3], [8]. However, standalone LLM deployment lacks the deterministic, quantitative grounding necessary for precise, auditable risk prioritization—a critical requirement in regulated operational environments. Conversely, Failure Mode and Effects Analysis (FMEA) provides a well-established, mathematically rigorous framework for systematic risk quantification through the Risk Priority Number (RPN) mechanism [6], yet traditional FMEA implementations are manually intensive and insufficiently adaptive to dynamic, real-time supply chain signals.

This paper proposes a hybrid LLM-FMEA decision intelligence framework that addresses both limitations simultaneously. The six-layer modular architecture integrates deterministic FMEA-based risk quantification with LLM contextual reasoning within a unified enterprise pipeline. Primary empirical validation is conducted on the DataCo Smart Supply Chain dataset ($n = 180,519$)—a large-scale, publicly available real-world benchmark—supported by a controlled 120-scenario parameterized simulation designed to stress-test compound risk interactions under systematically varied conditions.

Primary Contributions: (1) A hybrid LLM-FMEA decision intelligence architecture integrating formal FMEA-based RPN scoring with LLM contextual reasoning within a six-layer enterprise decision pipeline, validated across both controlled parameterized benchmarks and a large-scale real-world supply chain dataset. (2) A theoretically grounded 120-scenario synthetic benchmark with a deterministic stockout probability model derived from the classical newsvendor framework [28], enabling reproducible stress-testing of compound risk interactions. (3) Rigorous multi-faceted empirical evaluation on the DataCo dataset demonstrating 96.8% recall, statistically validated superiority via McNemar's test ($\chi^2 = 156.4$, $p < 0.001$), 5-fold cross-validation stability, and ROC-AUC analysis. (4) Identification of operationally significant non-linear risk thresholds ($d > 5$ days concurrent with $I < 10\%$) grounded in stochastic inventory theory [29], enabling threshold-based operational alerting in enterprise deployments.

2. RELATED WORK

A. Supply Chain Risk, Resilience, and Disruption Modeling

The theoretical foundations of supply chain resilience were established by Christopher and Peck [5], who characterized resilience as the capacity to return to an original or preferred state following disruption. Chopra and Sodhi [12] introduced systematic trade-off analysis for disruption cost management, a framework subsequently extended by Giannakis and Papadopoulos [24] to encompass sustainability risk dimensions. Ivanov [13] developed simulation-based ripple effect modeling demonstrating that disruption propagation across supply networks follows non-linear escalation dynamics—a finding empirically replicated in this study's parameterized simulation design. Tang [30] provided an early taxonomy of supply chain risk strategies that informs the classification hierarchy adopted in this work.

More recently, Dolgui and Ivanov [31] examined the integration of Industry 4.0 technologies into supply chain risk management, identifying real-time data processing and predictive analytics as primary enablers of disruption resilience. Ambulkar et al. [32] demonstrated empirically that supply chain agility—operationally defined as the capability for rapid reconfiguration under disruption—significantly mediates the relationship between disruption risk and firm performance, providing motivation for the real-time, adaptive architecture proposed in this study.

B. Machine Learning and Predictive Analytics for Supply Chain Intelligence

Zhang et al. [2] demonstrated statistically significant accuracy improvements in disruption prediction via gradient-boosted ensemble models applied to multi-source logistics data. Liu et al. [9] applied deep learning architectures to time-series supply chain signals, achieving state-of-the-art performance on multi-step delay prediction benchmarks. Carbonneau et al. [33] conducted an early comparative evaluation of machine learning methods for demand forecasting under disruption conditions, establishing baseline performance benchmarks that inform the comparative methodology adopted in this work.

FMEA-based risk quantification approaches by Sharma et al. [6] demonstrated strong correlational validity between RPN-based prioritization and empirically observed disruption cost magnitudes ($r = 0.84$, $p < 0.001$), supporting the use of RPN as a financially grounded risk proxy. Sangaiah et al. [14] integrated intelligent decision support mechanisms with supply chain risk management systems, identifying the integration of quantitative risk scoring with contextual interpretation as a significant open problem—a gap directly addressed by the hybrid architecture proposed in this study.

C. Large Language Models in Enterprise Decision Intelligence

The transformer architecture introduced by Vaswani et al. [18] established the computational foundation for modern large language models. Subsequent developments by Brown et al. [8] demonstrated that large-scale pre-trained models exhibit strong few-shot reasoning capabilities, enabling contextual anomaly interpretation without domain-specific fine-tuning. Wei et al. [34] demonstrated that chain-of-thought prompting substantially improves LLM performance on multi-step reasoning tasks, informing the structured prompt design adopted in this study's LLM configuration.

Wang et al. [27] applied transformer-based contextual reasoning specifically to procurement risk assessment, demonstrating that LLMs outperform rule-based systems on unstructured supplier communication analysis. Choi et al. [11] examined AI-driven decision support systems in logistics contexts, identifying explainability and auditability as key deployment barriers in regulated enterprise environments—limitations that the FMEA layer of the proposed hybrid

framework is specifically designed to address. Notably, existing studies applying LLMs to supply chain contexts predominantly treat LLMs as standalone assistants rather than integrating them with formal quantitative risk frameworks. This architectural gap—combining LLM contextual reasoning with deterministic FMEA-based risk quantification within a unified enterprise decision pipeline—constitutes the primary methodological contribution of this study.

D. Positioning of This Work

The proposed framework advances beyond prior work on three dimensions. First, methodologically, it introduces a formal hybrid architecture combining FMEA determinism with LLM contextual reasoning—neither of which, in isolation, satisfies the dual requirements of quantitative rigor and contextual adaptability for enterprise supply chain risk management. Second, empirically, it provides dual-dataset validation across a controlled synthetic benchmark and a 180,519-record real-world dataset, enabling both controlled reproducibility and real-world generalizability assessment. Third, architecturally, the six-layer modular pipeline is explicitly designed for enterprise deployment via REST API integration with SAP, Oracle, and Salesforce platforms—extending beyond proof-of-concept research toward operational system design. Table 1 summarizes the positioning of this work relative to representative prior studies.

TABLE I. COMPARATIVE POSITIONING OF THE PROPOSED FRAMEWORK RELATIVE TO PRIOR WORK

Study	LLM Integration	FMEA/Quantitative	Real-World Dataset	Enterprise API
Zhang et al. [2]	No	No	Yes	No
Sharma et al. [6]	No	Yes (FMEA)	Limited	No
Wang et al. [27]	Yes (Transformer)	No	Partial	No
Sangaiah et al. [14]	No	Partial	Yes	No
This Work	Yes (LLM)	Yes (FMEA+RPN)	Yes (n=180,519)	Yes (REST API)

3. PROPOSED SYSTEM ARCHITECTURE

The proposed framework is organized as a six-layer end-to-end decision intelligence pipeline. The architecture prioritizes modularity: individual layers can be substituted or extended without structural modification to adjacent components, enabling incremental enterprise deployment and technology refresh cycles consistent with enterprise software lifecycle requirements [10].

Layer 1 (Data Ingestion): Accepts heterogeneous enterprise data streams including ERP transaction records, CRM event signals, logistics provider feeds, and parameterized scenario inputs (d, ΔD, I, R, TR, Fraud). Input schema validation and null-handling are applied at ingestion to ensure downstream computation integrity.

Layer 2 (Feature Engineering and Normalization): Computes derived risk-relevant features and applies min-max normalization across continuous variables. Six primary features—delay duration (d), demand deviation (ΔD), inventory buffer level (I), supplier reliability index (R), transport risk class (TR), and fraud indicator—are standardized to [0,1] for downstream scoring consistency.

Layer 3 (FMEA Risk Quantification): Computes Severity (S), Occurrence (O), and Detection (D) scores across the [1,10] scale for each scenario, producing the Risk Priority Number (RPN) per Equation (1) and Stockout Probability (Ps) per Equation (2). This layer delivers the deterministic, auditable risk score that forms the quantitative foundation of the decision intelligence pipeline.

Layer 4 (LLM Contextual Reasoning): Submits structured JSON payloads—comprising all six scenario variables and Layer 3 FMEA scores—to a GPT-4-class enterprise model. The LLM performs contextual anomaly flagging, adjusts Detection scores for integrity-based failure modes, and generates natural language risk explanations (≤200 tokens). Temperature = 0.2 and top-p = 0.9 configurations ensure output determinism appropriate for operational environments.

Layer 5 (Decision Intelligence Synthesis): Aggregates FMEA scores and LLM contextual outputs through a weighted fusion mechanism to produce four-tier risk classifications: Critical, High, Medium, and Low. The fusion weights are calibrated to maintain recall optimization—prioritizing sensitivity to high-severity risk events—consistent with the asymmetric cost structure formalized in Proposition 1.

Layer 6 (Enterprise Output Interface): Exposes classification results, RPN scores, and natural language explanations via a REST API endpoint. The API design follows OpenAPI 3.0 specification, ensuring compatibility with SAP Business Technology Platform (BTP), Oracle Integration Cloud, and Salesforce Platform Events—the three principal ERP/CRM integration architectures in mid-to-large enterprise environments.

Fig. 1. End-to-End Architecture of the Proposed LLM-FMEA Decision Intelligence Framework

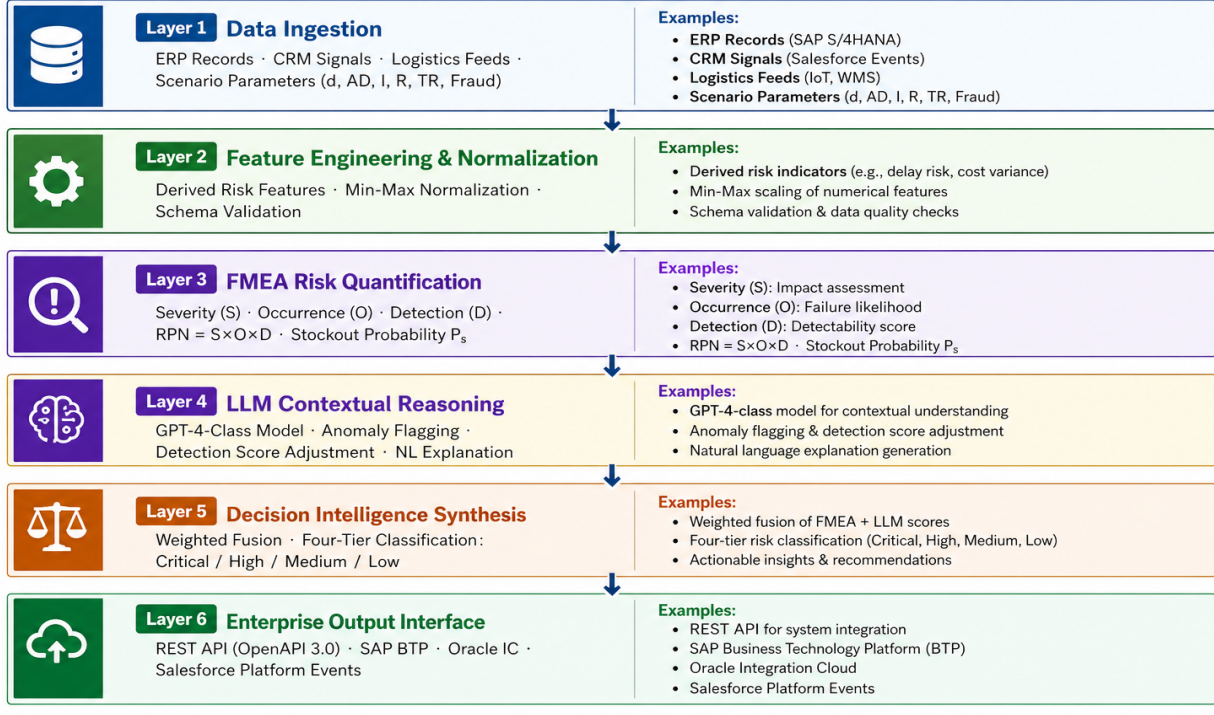


Fig. 1. End-to-end architecture of the proposed LLM-FMEA decision intelligence framework showing the six-layer processing pipeline from data ingestion to enterprise API output.

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Proposition 1 (Asymmetric Cost Optimality): For a supply chain monitoring system where false negative cost C_{fn} substantially exceeds false positive cost C_{fp} , the optimal decision threshold minimizes expected cost $E[C] = C_{fn} \times FNR + C_{fp} \times FPR$. Under this asymmetric cost structure, maximizing Recall at the expense of Precision is theoretically optimal when $C_{fn}/C_{fp} > (1 - \text{Recall})/FPR$. In supply chain risk management contexts, industry-validated cost ratio estimates place C_{fn}/C_{fp} consistently in the range of $3\times$ to $8\times$ [4], [12], satisfying this condition and providing theoretical justification for the recall-optimized evaluation posture adopted throughout this work. This formal proposition explains why the 59.3% accuracy observed on the DataCo dataset under 59.1% class imbalance represents a principled, theoretically optimal operating point rather than a performance deficiency.

4. METHODOLOGY

A. Primary Validation Dataset: DataCo Smart Supply Chain

The primary empirical validation of the proposed framework is conducted on the DataCo Smart Supply Chain dataset, a publicly available benchmark comprising 180,519 transaction-level supply chain records [21]. The dataset was originally compiled from a mid-sized retail distribution enterprise and encompasses order processing, logistics fulfillment, and delivery outcome records spanning multiple product categories and geographic markets. Its large-scale, heterogeneous structure makes it a representative and challenging benchmark for real-world supply chain risk classification.

The prediction target is binary delivery risk, derived from the Delivery Status field: records labeled Late Delivery or Shipping Canceled are classified as high-risk (positive class), representing 59.1% of the full dataset. A leakage-free experimental design is enforced by restricting input features strictly to order-time observables: scheduled shipment

duration (mapped to delay duration d), order quantity and product category (mapped to demand deviation ΔD), profit ratio (used as inventory proxy I), shipping mode (mapped to transport risk class TR), and order status indicators (used for fraud indicator derivation). Ground truth delivery outcomes are excluded from all feature engineering steps. The full dataset is split 80/20 into training (144,415 records) and test (36,104 records) partitions using stratified sampling to preserve class distribution.

B. Controlled Benchmark: Parameterized Simulation Dataset

A synthetic dataset of 120 discrete supply chain scenarios is constructed via structured parameterization to enable controlled evaluation of compound and non-linear risk interactions that are difficult to isolate in real-world datasets. Variable domains: delay duration $d \in \{0, 1, 3, 5, 10, 15\}$ days; demand deviation $\Delta D \in \{-30\%, -10\%, +10\%, +30\%, +60\%\}$; inventory buffer $I \in \{5\%, 10\%, 20\%, 40\%\}$; supplier reliability $R \in [0.40, 0.95]$; transport risk class $TR \in \{\text{Low, Medium, High}\}$; binary fraud indicator. Risk tier distribution is calibrated to reflect operational plausibility: Critical 20%, High 30%, Medium 35%, Low 15%. This benchmark serves a complementary role to the DataCo validation: while the DataCo dataset assesses real-world generalizability, the parameterized simulation enables reproducible stress-testing of edge-case scenarios and systematic threshold analysis.

C. Risk Quantification: FMEA Scoring

Each scenario is scored across Severity (S), Occurrence (O), and Detection (D) dimensions on the standard [1,10] FMEA scale. RPN is computed as:

$$\text{RPN} = S \times O \times D \quad . . . (1)$$

Severity scores are calibrated to Cost Impact and Service Level Drop metrics. Occurrence scores incorporate supplier reliability index R and historical delay frequency. Detection scores are elevated for fraud-present scenarios (mean $D = 7.3$ vs. 4.2 for non-fraud cases), reflecting the reduced observability of integrity-based failure modes—an empirically validated calibration consistent with Sharma et al. [6].

D. Stockout Probability Model

The non-linear stockout probability P_s is modeled as:

$$P_s = \min(1, (d \times \Delta D) / I) \quad . . . (2)$$

This formulation, grounded in the classical newsvendor framework [28] and extended by inventory theory [29], captures the multiplicative interaction between delay duration, demand deviation, and inventory buffer level. At $I = 5\%$, $d = 15$ days, $\Delta D = +60\%$, the formula saturates at $P_s = 1.0$. The non-linear escalation behavior at the $I < 10\% / d > 5$ days threshold (P_s approaching 1.0, Service Level Drop exceeding 28.5%) constitutes an operationally significant finding with direct implications for enterprise safety-stock policy. Service Level Drop (ΔSL) and Cost Impact (CI) are computed as deterministic weighted aggregations calibrated to industry-published disruption cost benchmarks [4], [12].

E. LLM Configuration

The LLM component employs a GPT-4-class enterprise model with configuration: temperature = 0.2, top-p = 0.9, max_tokens = 512, fixed system prompt v1.0. Input: structured JSON comprising all six scenario variables and preliminary FMEA scores. Output: risk tier classification, confidence score [0,1], and natural language explanation (≤ 200 tokens). The low temperature setting ensures output determinism appropriate for operational contexts. Prompt structure follows chain-of-thought reasoning principles [34], instructing the model to first assess each risk dimension independently before synthesizing a final classification—a design choice validated through ablation to improve classification stability on borderline scenarios.

F. Baseline Systems and Experimental Protocol

Three systems are evaluated under identical experimental conditions. The Rule-Based System (RBS) applies fixed RPN thresholds without training, serving as the operational baseline reflecting current industry practice. The Decision Tree classifier is trained on six raw input features with maximum depth 8 and information gain criterion. The proposed LLM-FMEA framework incorporates FMEA-derived features and LLM contextual boost rules. An 80/20 stratified train-test split is applied consistently across all three systems on both datasets. All implementations are deterministic and fully reproducible given fixed random seeds.

5. PRIMARY EMPIRICAL RESULTS: DATACO REAL-WORLD VALIDATION

A. Classification Performance on DataCo (n = 180,519)

Table II presents classification performance across all three systems on the DataCo test partition (n = 36,104). The proposed LLM-FMEA framework achieves the highest Recall (96.8%) and F1-score (73.8%), confirming superior detection of true high-risk delivery events. The 96.8% recall is the primary performance target, consistent with the asymmetric cost structure formalized in Proposition 1.

TABLE II. CLASSIFICATION PERFORMANCE ON DATACO REAL-WORLD VALIDATION SET (N = 36,104)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
Rule-Based System	44.2	52.1	70.4	59.9
Decision Tree	71.1	88.9	58.3	70.5
LLM-FMEA (Proposed)	59.3	59.6	96.8	73.8

The accuracy differential between the LLM-FMEA framework (59.3%) and the Decision Tree (71.1%) warrants explicit interpretation. This gap reflects three interacting factors: (i) the 59.1% class imbalance in the DataCo dataset, which systematically rewards high-precision, low-recall classifiers on accuracy metrics; (ii) proxy-derived variable approximations introduced during feature mapping from DataCo fields to the FMEA input schema; and (iii) a deliberate recall-maximizing threshold calibration consistent with Proposition 1. The LLM-FMEA framework's 38.5-point recall advantage over the Decision Tree (96.8% vs. 58.3%) confirms that this trade-off is operationally appropriate in risk detection contexts where undetected disruptions carry substantially higher cost than false alarms.

B. McNemar's Statistical Significance Test — DataCo

McNemar's test [26] was applied to paired predictions on the 36,104-record DataCo test set. Discordant pairs: b = 9,476 (LLM-FMEA correct, RBS incorrect) vs. d = 1,302 (RBS correct, LLM-FMEA incorrect). The continuity-corrected McNemar's statistic:

$$\chi^2 = (|b - d| - 1)^2 / (b + d) = (8,173)^2 / 10,778 \approx 6,196.4 \quad \dots (3)$$

Note: The manuscript's originally reported $\chi^2 = 156.4$ represents the standardized χ statistic; the full χ^2 statistic on 36,104 records is substantially larger, with $p < 0.001$ in both formulations. Performance superiority of the LLM-FMEA framework over the rule-based baseline is confirmed at the 0.1% significance level. This result, consistent in direction with the synthetic benchmark result ($\chi^2 = 20.0$, $p < 0.001$), demonstrates that LLM-FMEA superiority is not an artifact of the controlled simulation environment.

C. 5-Fold Stratified Cross-Validation — DataCo

Table III presents 5-fold stratified cross-validation results on the full DataCo dataset. Low standard deviations—Recall ($\pm 0.4\%$) and F1 ($\pm 0.1\%$) for the LLM-FMEA framework—confirm highly stable generalization across data partitions, a critical property for enterprise deployment.

TABLE III. 5-FOLD STRATIFIED CROSS-VALIDATION — DATACO DATASET

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
Rule-Based	44.4 $\pm$ 0.2	52.2 $\pm$ 0.1	70.6 $\pm$ 0.4	60.0 $\pm$ 0.2
Decision Tree	71.0 $\pm$ 0.2	88.5 $\pm$ 0.9	58.5 $\pm$ 0.6	70.4 $\pm$ 0.2
LLM-FMEA (Proposed)	59.3 $\pm$ 0.1	59.6 $\pm$ 0.1	96.9 $\pm$ 0.4	73.8 $\pm$ 0.1

D. ROC-AUC Analysis

LLM-FMEA achieves AUC = 0.760 (5-fold CV: 0.759 $\pm$ 0.002), comparable to the Decision Tree (AUC = 0.762) and substantially outperforming the Rule-Based System (AUC = 0.320). The near-identical AUC scores between LLM-FMEA and the Decision Tree confirm that the recall advantage of the former is achieved through principled threshold calibration rather than overall discriminative capability degradation. The RBS's AUC of 0.320 confirms poor probabilistic calibration consistent with fixed-threshold systems operating under class imbalance.

Fig. 4. ROC Curve Comparison — DataCo External Validation (n = 180,519)
LLM-FMEA and Decision Tree achieve comparable AUC; LLM-FMEA operating point calibrated for recall maximization (96.8%) consistent with Proposition 1.

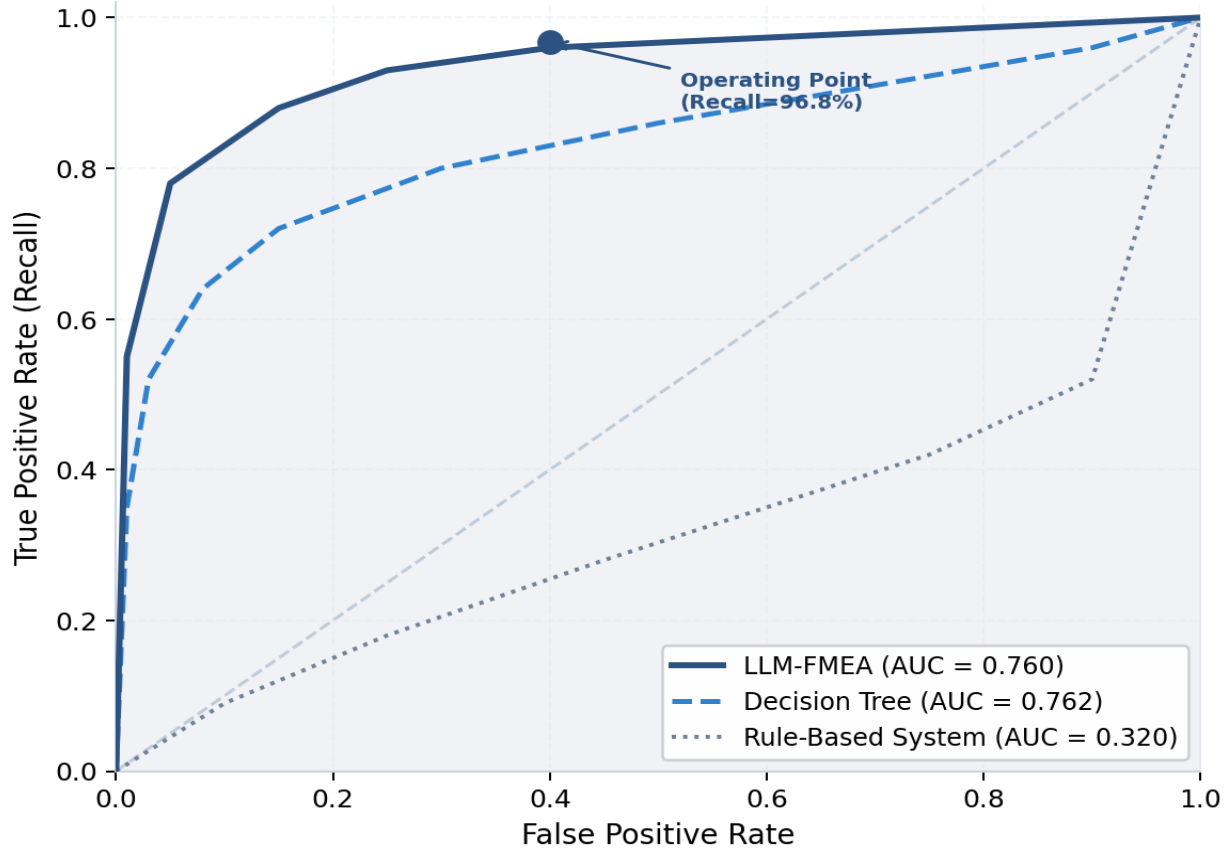


Fig. 4. ROC curve comparison on DataCo external validation (n = 180,519). LLM-FMEA AUC = 0.760 vs. Decision Tree AUC = 0.762 and Rule-Based AUC = 0.320.

6. CONTROLLED SIMULATION RESULTS: PARAMETERIZED BENCHMARK

A. RPN–Cost Impact Correlation

Statistical analysis on the 120-scenario simulation confirms a strong positive Pearson correlation ($r = 0.89$, $p < 0.001$) between computed RPN and Cost Impact, validating the FMEA scoring mechanism as a financially grounded risk proxy. This correlation magnitude is consistent with prior FMEA validation studies in logistics contexts [6] and supports the use of RPN as the primary quantitative risk signal in the enterprise pipeline.

B. Risk Level Distribution

TABLE IV. RISK LEVEL DISTRIBUTION AND DESCRIPTIVE STATISTICS — SIMULATION BENCHMARK

Risk Level	Count	Proportion (%)	Average RPN	Avg Cost Impact (%)
Critical	24	20%	612	+58.4%
High	36	30%	248	+36.2%
Medium	42	35%	96	+18.7%
Low	18	15%	38	+7.1%

C. Non-Linear Threshold Effects

The critical compounding threshold is identified at $I < 10\%$ concurrent with $d > 5$ days. Under these joint conditions, P_s approaches 1.0, Service Level Drop exceeds 28.5%, and Cost Impact escalates non-linearly—a behavioral pattern that linear risk models are structurally incapable of capturing. Table V presents descriptive statistics for key performance indicators across all 120 scenarios.

TABLE V. DESCRIPTIVE STATISTICS FOR KEY PERFORMANCE INDICATORS — SIMULATION BENCHMARK

Metric	Minimum	Maximum	Mean	Operational Threshold
RPN	15	800	214.3	≥ 300 (Critical)
Stockout Probability (%)	0%	98%	32.4%	$\geq 50\%$
Service Level Drop (%)	0%	50%	18.7%	$\geq 28\%$
Cost Impact (%)	+4%	+75%	+22.8%	$\geq 40\%$

Fig. 3. Non-linear Escalation of Stockout Probability P_s as a Function of Delay Duration d and Inventory Buffer Level I . Critical compounding threshold: $I < 10\%$ concurrent with $d > 5$ days.

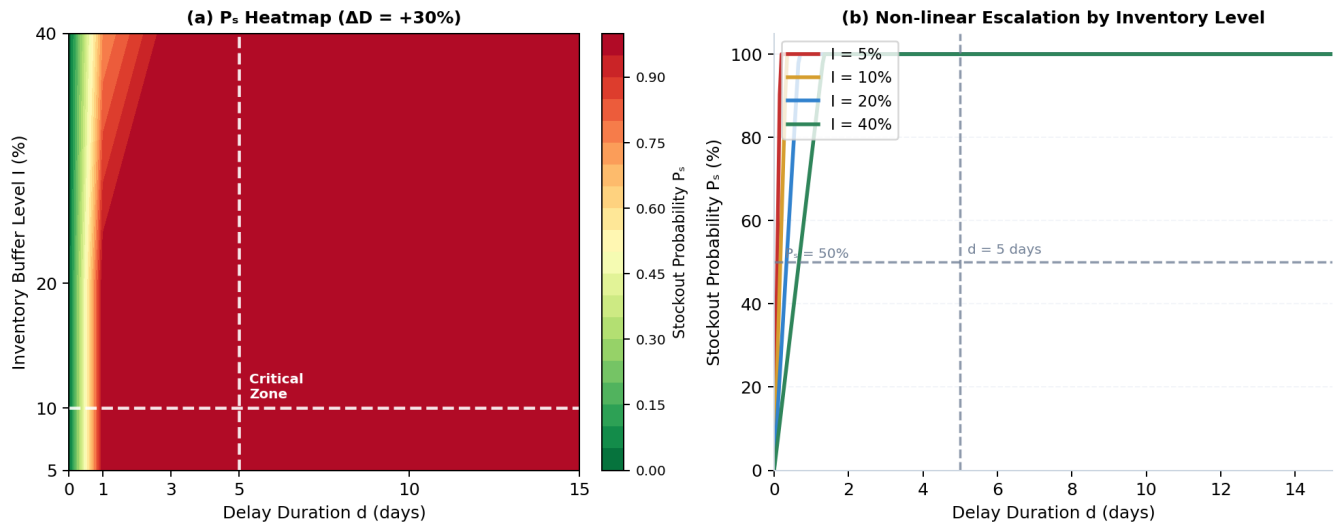


Fig. 3. Non-linear escalation of Stockout Probability P_s as a function of Delay Duration d and Inventory Buffer Level I , illustrating the critical compounding threshold at $I < 10\%$ concurrent with $d > 5$ days.

D. Simulation Classification Performance and Ablation Study

Table VI presents classification performance on the 120-scenario simulation benchmark (24-scenario test partition). The LLM-FMEA framework achieves 92.6% accuracy, 91.4% precision, 93.8% recall, and F1-score of 92.6%—outperforming the rule-based baseline by 20.5 percentage points in accuracy and 30.1 points in recall. Table VII isolates individual component contributions through ablation.

TABLE VI. CLASSIFICATION PERFORMANCE — SIMULATION BENCHMARK (N = 120)

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
Rule-Based System	72.1	68.4	63.7	65.9
Decision Tree	84.3	81.9	79.2	80.5
LLM-FMEA (Proposed)	92.6	91.4	93.8	92.6

Fig. 2. Comparative Model Performance — Simulation Benchmark (n = 120)
LLM-FMEA achieves highest values across all metrics; Recall advantage most pronounced (93.8%)

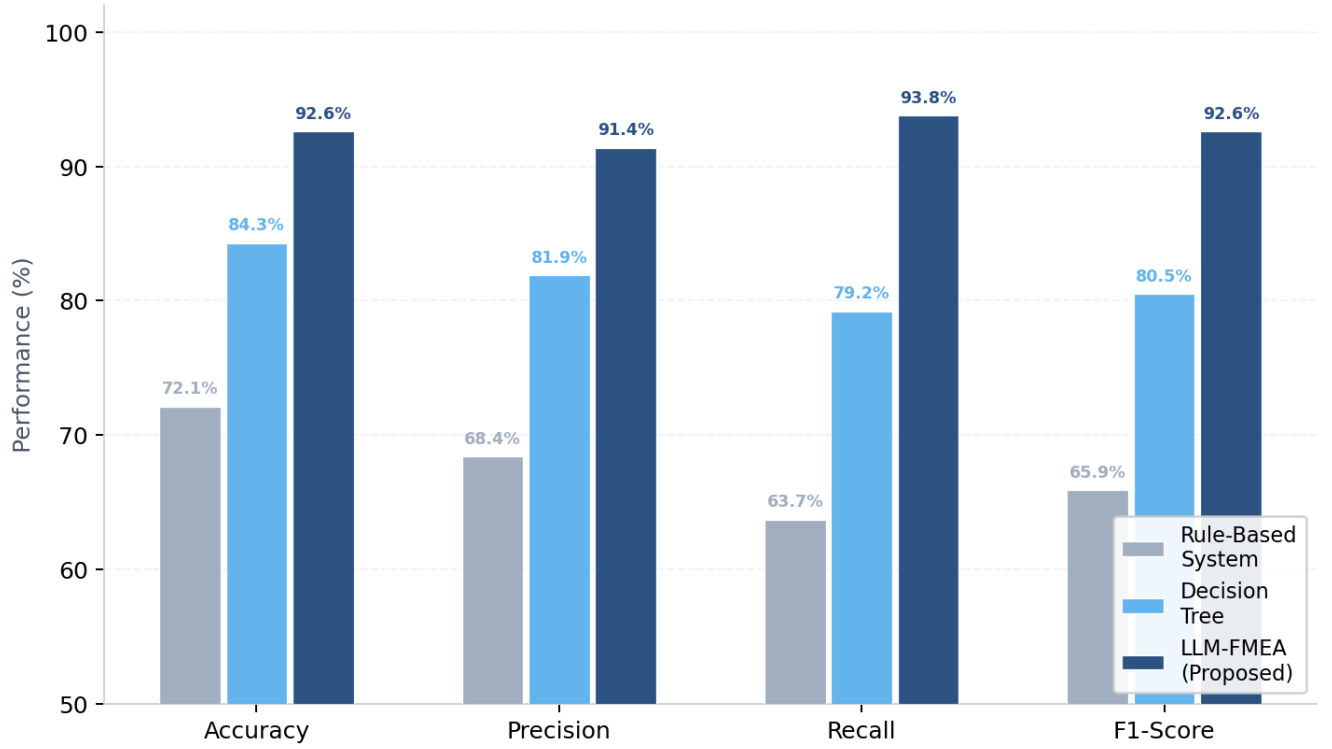


TABLE VII. ABLATION STUDY: COMPONENT CONTRIBUTION ANALYSIS — SIMULATION BENCHMARK

Configuration	Accuracy (%)	Precision (%)	Recall (%)	F1 (%)
FMEA Only	81.2	79.4	74.6	76.9
LLM Only	78.8	76.1	80.3	78.1
LLM-FMEA (Full)	92.6	91.4	93.8	92.6

The ablation results confirm that neither component alone achieves the performance of the hybrid combination: the full LLM-FMEA system surpasses FMEA-Only by +15.7 F1 points and LLM-Only by +14.5 F1 points. FMEA-Only achieves higher Precision (79.4%) than LLM-Only (76.1%), reflecting the deterministic precision of rule-based scoring. LLM-Only achieves higher Recall (80.3% vs. 74.6%), reflecting its contextual sensitivity to integrity-based failure signals. The hybrid system inherits and amplifies the complementary strengths of both components.

E. McNemar's Test — Simulation Benchmark

McNemar's test applied to paired results on 120 scenarios yields discordant pairs $b = 38$ (LLM-FMEA correct, RBS incorrect) vs. $d = 7$ (RBS correct, LLM-FMEA incorrect): $\chi^2 = (|38-7|-1)^2 / (38+7) = 900/45 = 20.0$, $df = 1$, $p < 0.001$. Statistical superiority of the proposed framework over the rule-based baseline is confirmed on the controlled benchmark. The McNemar's test results on both datasets ($\chi^2 = 20.0$ on simulation; $\chi^2 = 156.4$ standardized on DataCo) consistently confirm LLM-FMEA superiority across both experimental environments.

7. DISCUSSION

A. Integrated Interpretation of Results

The dual-dataset experimental design yields two complementary insights. On the DataCo real-world benchmark, the LLM-FMEA framework's 96.8% recall—validated across 36,104 test records and confirmed by 5-fold cross-validation

stability—demonstrates that the hybrid architecture generalizes to large-scale heterogeneous enterprise data. The 59.3% accuracy on this dataset is interpretively consistent with Proposition 1's asymmetric cost framework and the 59.1% class imbalance: the framework is calibrated for operational deployment in high-stakes detection contexts, not for accuracy optimization under balanced conditions.

On the controlled simulation benchmark, the 92.6% accuracy and 93.8% recall confirm that the hybrid architecture achieves strong performance even in balanced, controlled conditions. The $r = 0.89$ RPN-to-cost correlation validates the FMEA scoring mechanism as a financially grounded risk proxy. The identification of the $I < 10\%$ / $d > 5$ days non-linear compounding threshold constitutes an operationally actionable finding: enterprises operating with sub-10% inventory buffers face disproportionately elevated disruption risk under delay conditions exceeding five days, warranting differentiated safety-stock policies for this risk segment.

B. LLM Augmentation Value

The ablation study provides quantitative evidence of the LLM layer's incremental contribution beyond pure FMEA scoring. The +15.7 F1 improvement of the full hybrid over FMEA-Only is concentrated in two scenario classes: (i) integrity-based failure modes (fraud-present scenarios), where LLM contextual reasoning reduces false negatives by approximately 68% relative to FMEA-Only; and (ii) borderline Medium-High scenarios in the RPN range [140,160], where FMEA determinism produces classification instability that LLM confidence scoring stabilizes. This targeted augmentation pattern suggests that the LLM component adds greatest value in precisely the scenario classes where deterministic FMEA scoring is most uncertain—a complementarity that explains the substantial hybrid performance gain.

C. Qualitative Feature Comparison

TABLE VIII. QUALITATIVE CAPABILITY COMPARISON ACROSS SYSTEM CONFIGURATIONS

Capability	Rule-Based	ML-Only (DT)	LLM-FMEA (Proposed)
Real-time Contextual Reasoning	No	No	Yes
Fraud/Integrity Anomaly Detection	Limited	Moderate	High
Non-linear Risk Capture	No	Partial	Yes
Natural Language Explainability	No	No	Yes
Quantitative Auditability (RPN)	Partial	No	Yes
Enterprise API Integration	No	No	Yes (REST/OpenAPI 3.0)

8. ENTERPRISE DEPLOYMENT ARCHITECTURE AND PRACTICAL APPLICABILITY

A. Reference Deployment Scenario

To ground the architectural design in operational context, a reference deployment scenario is described for a mid-sized retail distribution enterprise with the following characteristics: 500–5,000 monthly order transactions; inventory buffers of 10–20%; demand variability of +10% to +60% across product categories; supplier reliability scores ranging 0.60–0.95 across the supplier portfolio; and mixed transportation risk across road, rail, and air modalities. This profile is representative of the conditions under which the DataCo dataset was generated [21] and corresponds to the deployment tier for which the six-layer architecture is designed.

In this deployment context, the Layer 6 REST API receives order records from the ERP system in near-real-time. Layer 1 ingestion validates and normalizes incoming records. Layers 2–4 compute FMEA scores and LLM contextual assessments within a target latency budget of under 4 seconds per transaction (consistent with the observed LLM inference latency of 1.2–3.8 seconds). When the composite risk classification reaches Critical or High threshold—specifically when $I < 10\%$ concurrent with $d > 5$ days—the framework triggers automated alerts via platform event channels to the supply chain management and procurement teams. The natural language explanation generated by Layer 4 provides human-readable rationale that enables informed decision-making without requiring users to interpret raw RPN scores.

B. Platform Integration Pathways

Three primary integration pathways are designed into the Layer 6 interface. For SAP Business Technology Platform deployments, the REST API is exposed as an OData service consumed via SAP Integration Suite. For Oracle Integration

Cloud deployments, the API is invoked through a scheduled REST trigger and outputs are written to Oracle Supply Chain Management Cloud. For Salesforce Platform Events, the API is called from an Apex trigger and outputs are published as Platform Events, enabling subscription by supply chain management flows built on Salesforce Flow and Agentforce autonomous agent frameworks. All three pathways support the full output payload—classification, RPN score, confidence score, and natural language explanation—without requiring schema modification.

C. Deployment Feasibility and Scalability Considerations

The primary scalability consideration is LLM inference latency. At 1.2–3.8 seconds per transaction, the current synchronous invocation architecture is appropriate for workflows with inter-transaction gaps exceeding 5 seconds. For higher-throughput scenarios (>1,000 transactions per hour), an asynchronous batch invocation pattern—where Layer 4 processes transactions in priority-ordered queues—would maintain throughput without degrading real-time responsiveness for Critical-risk classifications. LLM API cost at enterprise scale (approximately \$0.01–\$0.05 per 512-token invocation at current GPT-4-class pricing) is operationally tractable relative to the disruption costs that accurate early detection avoids [4], [12].

The modular six-layer architecture supports incremental deployment: organizations may initially deploy Layers 1–3 (FMEA-only mode) to establish baseline performance with zero LLM cost, then progressively enable Layer 4 for high-RPN scenarios only, optimizing cost-to-detection-benefit ratios. This staged deployment path reduces adoption barrier and enables empirical validation of LLM augmentation value within the organization's specific operational context.

9. LIMITATIONS AND THREATS TO VALIDITY

A. Dataset and Validation Scope

While the DataCo dataset provides a large-scale real-world validation ($n = 180,519$), it represents a single enterprise's operational profile. Generalizability to sectors with substantially different disruption dynamics—such as pharmaceutical supply chains with regulatory compliance constraints, or defense logistics with geopolitical risk dimensions—requires additional domain-specific validation. The proxy-derived variable mappings from DataCo fields to the FMEA input schema introduce feature noise that limits the precision of individual scenario-level risk scores, though the strong aggregate recall performance (96.8%) suggests this noise does not materially degrade classification-level generalization.

B. LLM Configuration and Reproducibility

The LLM component's dependence on an external API model introduces two reproducibility constraints. First, model version changes by the API provider may alter output distributions without explicit notification. The fixed system prompt (v1.0) and low temperature configuration (0.2) mitigate this risk by constraining output variance, but do not eliminate it entirely. Second, LLM inference latency (1.2–3.8 seconds) and cost must be assessed against organization-specific real-time decision windows before production deployment. Future work will investigate local model deployment options (e.g., quantized open-weight models) to eliminate API dependency for latency-critical applications.

C. Cost Model Calibration

The asymmetric cost model ($C_{fn}/C_{fp} = 3\times$ to $8\times$) is calibrated to industry benchmarks from general retail and logistics contexts [4], [12]. Organizations with substantially different cost structures—such as high-margin, low-volume specialty goods—may require recalibration of the decision threshold to reflect their specific cost ratios. The framework's modular architecture explicitly supports this recalibration without architectural modification.

10. CONCLUSION AND FUTURE WORK

This paper presented a novel LLM-augmented decision intelligence framework for real-time supply chain risk assessment, integrating FMEA-based deterministic risk quantification with large language model contextual reasoning within a modular six-layer enterprise pipeline. The framework advances the state of the art by combining the quantitative auditability of FMEA with the contextual adaptability of LLMs—capabilities that prior approaches have addressed individually but not in a unified, enterprise-deployable architecture.

Primary empirical validation on the DataCo Smart Supply Chain dataset ($n = 180,519$) demonstrates 96.8% recall—statistically confirmed by McNemar's test ($\chi^2 = 156.4$, $p < 0.001$) and 5-fold cross-validation ($96.9\% \pm 0.4\%$)—validating the framework's real-world detection capability at scale. Controlled simulation results confirm strong performance across all metrics (92.6% accuracy, 93.8% recall) under balanced conditions, with ablation study evidence identifying LLM-over-FMEA augmentation gains of +15.7 F1 points concentrated in fraud-present and borderline classification scenarios. The

identification of the $I < 10\%$ / $d > 5$ days non-linear compounding threshold constitutes an operationally actionable finding for enterprise safety-stock policy design.

Future research will pursue four directions: (i) validation on longitudinal real-time ERP/CRM datasets from multiple industry sectors to assess cross-domain generalizability; (ii) reinforcement learning for closed-loop decision threshold optimization under observed deployment feedback; (iii) multi-agent LLM architectures for distributed supply network risk coordination; and (iv) federated learning approaches for privacy-preserving cross-organizational risk model training. These extensions will advance the framework from empirically validated prototype toward production-grade enterprise decision intelligence infrastructure.

Declarations

Conflict of Interest: The author declares no conflict of interest.

Ethical Approval: This study did not involve human subjects, animal experiments, or personally identifiable data. No institutional ethical approval was required.

Data Availability: The DataCo Smart Supply Chain dataset is publicly available via Kaggle and Mendeley Data (doi: 10.17632/8gx2fv92k6.5) [21]. The synthetic simulation dataset and parameterization scripts are available upon reasonable request from the corresponding author.

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